

This database design focuses on ACID compliance, auditability, and scalability to meet the requirements of a Banking Management System.

1. Database Overview

The system uses a relational schema to ensure transactional integrity (crucial for ledger balances). It separates user identity, account management, transactional data, and loan processing into distinct domains. Audit logs are centralized to satisfy compliance and fraud detection requirements.

2. ER Diagram Description

- **Users** are the core entity, linked to **Roles** (RBAC).
- **Accounts** belong to **Users**.
- **Transactions** link two **Accounts** (Internal) or one **Account** to an external entity.
- **LoanApplications** are linked to **Users** and contain a foreign key to the **Employee** who processed them.
- **AuditLogs** and **FraudAlerts** track system and user activity.

3. Tables & Columns

- **Users**: `user\_id`, `email`, `password\_hash`, `mfa\_secret`, `last\_password\_change`, `status`
- **Accounts**: `account\_id`, `user\_id`, `account\_type`, `balance`, `created\_at`
- **Transactions**: `tx\_id`, `source\_acc\_id`, `dest\_acc\_id`, `amount`, `tx\_type`, `status`, `timestamp`
- **LoanApplications**: `loan\_id`, `user\_id`, `amount`, `status`, `processed\_by`, `updated\_at`
- **AuditLogs**: `log\_id`, `user\_id`, `action`, `ip\_address`, `timestamp`
- **FraudAlerts**: `alert\_id`, `tx\_id`, `severity`, `status`, `resolved\_by`

4. Relationships & Keys

- **Primary Keys (PK)**: Surrogate IDs (UUIDs or BigInt) for all tables.
- **Foreign Keys (FK)**:

* `Accounts.user\_id` -> `Users.user\_id`

* `Transactions.source\_acc\_id` -> `Accounts.account\_id`

* `LoanApplications.user\_id` -> `Users.user\_id`

* `LoanApplications.processed\_by` -> `Users.user\_id` (Internal Employee)

5. SQL Schema (PostgreSQL Dialect)

```
CREATE TABLE Users (  
    user_id UUID PRIMARY KEY,  
    email VARCHAR(255) UNIQUE NOT NULL,  
    password_hash TEXT NOT NULL,  
    mfa_secret TEXT NOT NULL,  
    last_password_change TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    role VARCHAR(50) DEFAULT 'CUSTOMER'  
);  
  
CREATE TABLE Accounts (  
    account_id UUID PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    balance DECIMAL(19,4) NOT NULL DEFAULT 0.00,  
    account_type VARCHAR(50),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE Transactions (  
    tx_id UUID PRIMARY KEY,  
    source_acc_id UUID REFERENCES Accounts(account_id),  
    dest_acc_id UUID,  
    amount DECIMAL(19,4) NOT NULL,  
    tx_type VARCHAR(50),  
    status VARCHAR(20),  
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE LoanApplications (  
    loan_id UUID PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    amount DECIMAL(19,4) NOT NULL,  
    status VARCHAR(20) DEFAULT 'PENDING',  
    processed_by UUID REFERENCES Users(user_id),  
    updated_at TIMESTAMP  
);  
  
CREATE TABLE AuditLogs (  
    log_id SERIAL PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    action TEXT,  
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

6. Indexes & Constraints

- ****Indexes****:
 - * ``idx_tx_timestamp``: For time-series analysis of spending.
 - * ``idx_acc_user``: For fast lookup of user balances.
 - * ``idx_loan_status``: For processing dashboards.

- **Constraints**:
 - * `CHECK (balance >= 0)`: Enforced on savings/checking accounts.
 - * `CHECK (amount > 0)`: For all transactions.
 - * **90-Day Policy**: Application logic triggers password change prompt if `CURRENT_DATE - last_password_change > 90`.

7. Normalization Notes

- **3NF**: The schema is normalized to Third Normal Form. Transactional logic separates the "sender" and "receiver" accounts into the `Transactions` table to ensure atomic balance updates.
- **Denormalization Consideration**: The `balance` field in `Accounts` is kept as a summary, but all actual financial movements are calculated via the `Transactions` log (Ledger approach) to ensure audit accuracy required by banking regulations.
- **Data Integrity**: Used `DECIMAL(19,4)` for currency to prevent floating-point rounding errors common in banking applications.